| Week | Phase & Detailed Tasks | Objective & Core Purpose | Est. Time & Diff. | Success Criteria (10/10) |
|---|---|---|---|---|
| **W1** | **Phase 1: Env Setup & Reproducing Baselines**<br>1. Setup virtual environment with PyTorch 2.3+ & CUDA 12.<br>2. Download pre-trained weights for SD 1.5, SDXL, SDXL-Lightning, LCM-LoRA, and BLIP-2.<br>3. Execute and baseline SemanticDraw on RTX 2080 Ti/3090. | To establish a stable development sandbox and verify that the base models replicate the published latency (e.g., 1.57 FPS) and visual quality before inserting the proposed modules. | 45 hrs<br>Moderate | - Baseline reproduces official paper's performance (1.57 FPS on single RTX 2080 Ti).<br>- Establish a clean, modular API codebase to ease subsequent U-Net modifications. |
| **W2** | **Phase 1: Bilateral Masking Implementation**<br>1. Instrument cross-attention hooks in U-Net to extract attention map $A_t = \mathrm{CrossAttn}(x_t, y)$.<br>2. Locate semantic anchor $I = \arg\max_{p \in M} A_t(p)$.<br>3. Calculate spatial penalty $P_{\mathrm{space}}$ and semantic penalty $P_{\mathrm{sem}}$ using PyTorch tensor operations on GPU. | To replace passive, blind quantized masks (which cause feature leakage and color blending) with an active, object-aware bilateral mask that respects semantic boundaries. | 55 hrs<br>Hard | - **Zero CPU Bottleneck:** Complete elimination of nested `for` loops at the pixel level.<br>- Bilateral tensor computation runs in under 5ms, preserving sub-second interactive requirements. |
| **W3** | **Phase 1: Symbiotic Pipeline Integration**<br>1. Build a dual-model wrapper to coordinate Student $\theta$ (SDXL-Lightning) and Teacher $\psi$ (SDXL) simultaneously.<br>2. Implement step-wise Tweedie denoising $\hat{x}_0^{\theta}$ and renoising to $s = t - 1$.<br>3. Code interpolative consensus $\hat{x}_0^{\mathrm{new}} = (1-\lambda)\hat{x}_0^{\theta} + \lambda \hat{x}_0^{\psi}$. | To bridge the quality and structural gap of few-step consistency models by applying teacher-guided (SDXL) refinement during the early sampling stages without retraining. | 65 hrs<br>**Extremely Hard** | - **No Out-Of-Memory (OOM) Errors:** Stable execution on a single 24GB GPU using FP16 and gradient checkpointing.<br>- Mathematical trajectory of the student model is actively corrected by the teacher model. |

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| **W4** | **Phase 1: System Integration & PoC**<br>1. Substitute Latent Pre-Averaging with active Weighted Averaging using $W_t$.<br>2. Incorporate white-background bootstrapping and coordinate-based mask-centering.<br>3. Generate qualitative samples on high-contrast overlaps (e.g., “Fire” vs. “Water”). | To unify all algorithmic components into a single, cohesive generation loop and verify the baseline improvements. | 50 hrs<br>Hard | **[MILESTONE 1 - POC VALIDATED]**<br>Visual inspection confirms that high-contrast semantic boundaries exhibit sharp edges without color blending, artifacts, or bleeding. |
| **W5** | **Phase 2: Sub-Second Real-Time Acceleration**<br>1. Integrate Tiny AutoEncoder (TAESD) for high-speed latent decoding.<br>2. Implement asynchronous caching for CLIP text embeddings.<br>3. Profile exact execution bottlenecks (U-Net forward, bilateral blending, renoising). | To optimize the pipeline throughput, ensuring the dual-model computation does not break the sub-second interactive “semantic palette” requirement. | 45 hrs<br>Moderate | - Achieve a real-world generation speed of **over 1.5 FPS** on an RTX 2080 Ti.<br>- Demonstrate that the additional teacher evaluation step introduces less than 5% latency overhead due to tensor batching. |
| **W6** | **Phase 2: Hyperparameter Grid Search**<br>1. Automate grid search across spatial variance $\sigma_D$ and semantic variance $\sigma_S$.<br>2. Assess optimal mask-centering steps (test range 1–4).<br>3. Export high-resolution parameter sensitivity curves. | To locate the optimal parameter “sweet spot” that maximizes object structural harmony and mask boundary adherence. | 40 hrs<br>Moderate | - Pinpoint the exact “sweet spot” (e.g. mask-centering steps = 2) for maximum overall harmony and mask adherence.<br>- Export high-resolution vector graphs illustrating the grid-search behavior. |

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| **W7** | **Phase 2: Theoretical Formulation**<br>1. Formalize Section 3 (Methodology) in LaTeX.<br>2. Express the symbiotic distillation process as a proximal optimization on the SDS loss.<br>3. Derive the mathematical proof of bypassing the score Jacobian via Decomposed Diffusion Sampling (DDS). | To solidify the academic narrative and prove to CVPR reviewers that our framework is backed by rigorous mathematical foundations rather than empirical heuristics. | 35 hrs<br>Hard | - Absolute mathematical consistency with foundational diffusion works (DDIM, Score SDE, DDS).<br>- Reviewers are presented with a solid theoretical foundation rather than a collection of heuristical code. |
| **W8** | **Phase 3: COCO Dataset Pipeline Setup**<br>1. Extract 10,000 image-caption pairs from MS-COCO Validation.<br>2. Extract foreground object masks and labels as brush inputs, and captions as background.<br>3. Format masks using nearest-neighbor interpolation to standard $512 \times 512$ / $1024 \times 1024$ shapes. | To establish a standardized, unbiased evaluation benchmark that replicates real-world user drawing commands for large-scale quantitative testing. | 50 hrs<br>Moderate | - Scalable, robust batch processing script on the cluster that handles 10k items with automatic crash-recovery.<br>- Strict standardization of evaluation inputs to prevent bias. |
| **W9** | **Phase 3: Quantitative Benchmark Execution**<br>1. Generate 10k test images for proposed and baseline methods.<br>2. Compute FID, IS, CLIPfg (foreground alignment), and CLIPbg (background alignment) scores.<br>3. Evaluate samples with ImageReward and PickScore models. | To collect rigorous empirical data demonstrating that our method outperforms consistency baselines in image quality and semantic fidelity. | 45 hrs<br>Hard | **[MILESTONE 2 - SOTA PERFORMANCE ACHIEVED]**<br>- Proposed pipeline outperforms LCM baseline, reducing FID by 10–15%.<br>- ImageReward and PickScore demonstrate statistically significant improvements ($p$-value $< 0.05$). |

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| **W10** | **Phase 3: Cross-Baseline Comparison**<br>1. Run evaluations on MultiDiffusion (50 steps), MultiDiffusion+LCM (5 steps), LRDiff+LCM, and baseline SemanticDraw.<br>2. Track peak GPU memory (VRAM) usage and wall-clock times.<br>3. Compile a comprehensive, clean comparative table. | To verify that our method provides the best speed-to-quality trade-off, matching the quality of slow samplers (50 steps) at few-step speeds. | 50 hrs<br>Moderate | - Complete a comprehensive quantitative table (Table 1) proving our method achieves DDIM-quality output (50 steps) at few-step speeds.<br>- Unfair parameter tuning of competitor baselines is strictly avoided. |
| **W11** | **Phase 3: User Preference & Ablation Studies**<br>1. Execute double-blind A/B testing user study with 30 participants (including graphics professionals).<br>2. Conduct ablation tests (remove bilateral filter, swap anchor $I$ for geometric center, remove Distillation++).<br>3. Perform statistical significance testing ($p$-value). | To validate human preference for our approach and isolate the specific performance contributions of each algorithmic component. | 45 hrs<br>Hard | - User Study complies with human-subject research standards; report preference ratios clearly.<br>- Ablation results prove each module is non-redundant and crucial to structural harmony. |
| **W12** | **Phase 4: Architecture Diagram & Teaser**<br>1. Design Figure 2 (Block diagram) detailing Student/Teacher tensor interactions and bilateral blending.<br>2. Capture Figure 1 (Teaser) showcasing the sub-second drawing stream.<br>3. Visualise attention heatmap dynamics. | To create stunning visual assets that immediately hook the reviewers and convey the entire system's contribution in under 10 seconds of reading. | 40 hrs<br>Hard | - Diagrams look highly professional, with uniform typography, consistent colors, and clear labels.<br>- Teaser figure acts as a 10-second explanation of the entire paper's contribution. |

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| **W13** | **Phase 4: Qualitative Comparisons Layout**<br>1. Select the most visually striking comparative samples.<br>2. Create side-by-side grids with yellow zoom-in bounding boxes to highlight artifacts (e.g. limb deformities, color bleeding) and our corrections.<br>3. Format tables in LaTeX. | To provide undeniable qualitative evidence of our method's superiority in resolving anatomical distortions and edge bleeding. | 35 hrs<br>Moderate | - Breathtaking qualitative comparison layouts at native 1024x1024 resolution.<br>- Captions are self-explanatory and guide the reviewer's eyes immediately to our structural improvements. |
| **W14** | **Phase 4: Writing Introduction & Related Work**<br>1. Write a compelling Introduction framing the incompatibility of speed and control.<br>2. Characterize "feature leakage" as a fundamental mathematical loophole.<br>3. Draft a respectful and thorough Related Work section. | To establish a powerful, logically sound narrative that convinces the reviewers of the paper's significance and scientific contribution. | 40 hrs<br>Hard | - Meticulous academic writing that hooks the reader from paragraph one.<br>- Discloses a clear, convincing thesis on why this research represents a major milestone. |
| **W15** | **Phase 4: Experiments, Supplementary & Abstract**<br>1. Complete Section 4 (Experiments) with deep statistical insights.<br>2. Draft Supplementary Material explaining demo GUI, hyperparameters, and failure modes.<br>3. Refine Abstract (150 words) and Conclusion. | To compile the final manuscript draft and create a detailed supplementary package that ensures reproducibility and academic honesty. | 45 hrs<br>Moderate | - Draft v1 is complete, polished, and fully reviewed.<br>- Transparent discussion of limitations (e.g. shared latent space constraint) showcases academic integrity. |

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| **W16** | **Phase 5: Internal Review & CMT Submission**<br>1. Run internal review within the lab and incorporate advisor's edits.<br>2. Correct LaTeX errors, overfull/underfull hboxes, and clean BibTeX entries.<br>3. Zip clean source code and README.<br>4. Submit to CMT portal. | To perform rigorous quality assurance on the submission package and complete the submission ahead of the portal's peak traffic window. | 50 hrs<br>Hard | **[MILESTONE 3 - SUBMITTED TO CMT]**<br>- Submission completed 24 hours ahead of the deadline to avoid CMT portal network congestion.<br>- Standard-compliant PDF with no missing citations or references. |

Table 1: Plan for CVPR.